Name of the Course

Lecture 1: Topic of the lecture

Name of the Lecturer

Department of Materials Science, Montanuniversität Leoben, Austria

name.surname@unileoben.ac.at

Winter term 2026/27

Versetz' Berge

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Basics

1. Basics
2. Lists and overlays
3. Figures
4. More tools

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Text and highlighting

- important content is highlighted with `\alert{…}`
- less important content is highlighted with `\Alert{…}`
- The MUL display font (Funnel Display) is switched on with `\funneldisplay`; it is used for the frame titles
- mathematics works as usual: $a^2 + b^2 = c^2$, $\sin \alpha = \frac{b}{c}$

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An alert block

For warnings and important statements.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An alert block

For warnings and important statements.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An example block

For examples.

An alert block

For warnings and important statements.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An alert block

For warnings and important statements.

An example block

For examples.

A block without title.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Lists and overlays

1. Basics
2. Lists and overlays
3. Figures
4. More tools

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Lists

By default, list items are uncovered one by one (<+-> is the class default):

- first level items have squares

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Lists

By default, list items are uncovered one by one (<+-> is the class default):

- first level items have squares
  - lower levels have dashes

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Lists

By default, list items are uncovered one by one (<+-> is the class default):

- first level items have squares
  - lower levels have dashes
- *Example:* examples are marked with `\example` instead of `\item`

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Lists

By default, list items are uncovered one by one (<+-> is the class default):

- first level items have squares
  - lower levels have dashes
- *Example:* examples are marked with `\example` instead of `\item`
- last item

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Overlays

- `\only<2>{…}`: content exists on slide 2 only (takes no space otherwise)
- `\uncover<3->{…}`: space for the content is always reserved, the content is shown from slide 3 on

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Overlays

- `\only<2>{…}`: content exists on slide 2 only (takes no space otherwise)
- `\uncover<3->{…}`: space for the content is always reserved, the content is shown from slide 3 on

This text is shown with `\only<2>`.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Overlays

- `\only<2>{…}`: content exists on slide 2 only (takes no space otherwise)
- `\uncover<3->{…}`: space for the content is always reserved, the content is shown from slide 3 on

This text is shown with `\uncover<3->`.

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Figures

1. Basics
2. Lists and overlays
3. Figures
4. More tools

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

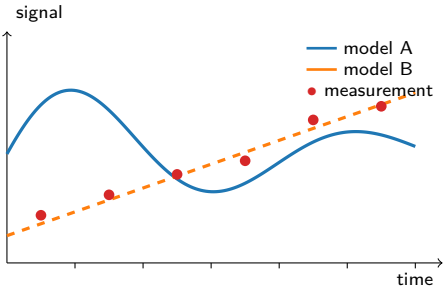
Figures and literature

- figures are kept in the folder `figs`, graphics (logos, background) in `graphics`
- images are included with `\includegraphics{figs/…}`; these demo figures are TikZ drawings included with `\input`
- `\literature{…}` (outside a frame) prints references above the footer of this and all following slides; clear it with `\literature{}`

A. Author, B. Author, and C. Author, *J. Example Sci.* **1**, 1 (2026)
D. Author and E. Author, *Example Lett.* **2**, 100 (2025)

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Figures and literature

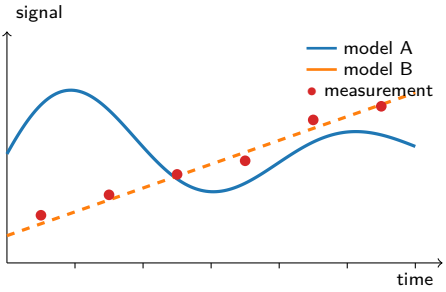


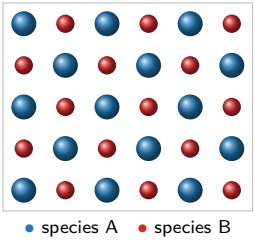
- figures are kept in the folder `figs`, graphics (logos, background) in `graphics`
- images are included with `\includegraphics{figs/…}`; these demo figures are TikZ drawings included with `\input`
- `\literature{…}` (outside a frame) prints references above the footer of this and all following slides; clear it with `\literature{}`

A. Author, B. Author, and C. Author, *J. Example Sci.* **1**, 1 (2026)
D. Author and E. Author, *Example Lett.* **2**, 100 (2025)

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Figures and literature





- figures are kept in the folder `figs`, graphics (logos, background) in `graphics`
- images are included with `\includegraphics{figs/…}`; these demo figures are TikZ drawings included with `\input`
- `\literature{…}` (outside a frame) prints references above the footer of this and all following slides; clear it with `\literature{}`

A. Author, B. Author, and C. Author, *J. Example Sci.* **1**, 1 (2026)
D. Author and E. Author, *Example Lett.* **2**, 100 (2025)

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Placing objects with `overdraw`

- the `overdraw` environment is a TikZ picture with absolute coordinates: one unit is 1/16 of the slide width, the origin is the bottom-left corner of the slide (right of the sidebar, if shown)
- `\framegrid` inside `overdraw` draws the coordinate grid for finding positions (slide 2) – remove it in the final version
- slide 3: label and arrow placed with `overdraw`

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

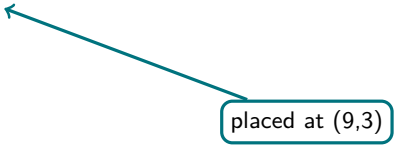
Placing objects with `overdraw`

- the `overdraw` environment is a TikZ picture with absolute coordinates: one unit is 1/16 of the slide width, the origin is the bottom-left corner of the slide (right of the sidebar, if shown)
- `\framegrid` inside `overdraw` draws the coordinate grid for finding positions (slide 2) – remove it in the final version
- slide 3: label and arrow placed with `overdraw`

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Placing objects with `overdraw`

- the `overdraw` environment is a TikZ picture with absolute coordinates: one unit is 1/16 of the slide width, the origin is the bottom-left corner of the slide (right of the sidebar, if shown)
- `\framegrid` inside `overdraw` draws the coordinate grid for finding positions (slide 2) – remove it in the final version
- slide 3: label and arrow placed with `overdraw`



Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

More tools

1. Basics
2. Lists and overlays
3. Figures
4. More tools

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Hidden results

- `\showResults{…}` shows its content with the class option `showResults` (default), and leaves an empty space of the same size with `hideResults`
- useful for exercises in lecture handouts

Exercise

Solve $x^2 - 5x + 6 = 0$.

Result: $x_1 = 2, \quad x_2 = 3$

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Frame logo

- the logo in the top-right corner can be changed with `\setFrameLogo{graphics/…}`
- here: the logo of the Department of Materials Science

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Backup

1. Basics
2. Lists and overlays
3. Figures
4. More tools

Basics
Text and highlighting
Blocks and columns
Lists and overlays
Lists
Overlays
Figures
Figures and literature
Placing objects
More tools
Hidden results
Frame logo
Backup

Backup slide

- slides after `\lastslide` are not counted in the total number of slides
- useful for backup slides